

**Sixth Semester B. Tech. (Electronics and Communication Engineering) Examination**

**INTRODUCTION TO INTERNET OF THINGS**

Time : 3 Hours ]

[ Max. Marks : 60

**Instructions to Candidates :—**

- (1) All questions are compulsory and carry marks as indicated.
- (2) Due credit will be given to neatness.
- (3) Assume suitable data and illustrate answers with neat sketches wherever necessary.

1. (A) For a E-health Use Case, recommend an architecture with details and justification. 7(CO1,5)  
(B) As an IoT Application Developer, select an Access Technology for Swarm Drone applications. Justify. 3(CO1,5)
2. (A) Compare Raspberry pi and ESP8266 Node MCU. 4(CO1)  
(B) Describe the steps to enable I2C interface on Rpi using terminal and through GUI. 4(CO2)
3. (A) Write a bash script to read names from a file “names.txt” and display the names using loop with the following message “Hello <name from the file>, How are you ? 6(CO2)  
(B) Write a bash script to execute a python source file “test.py” with two input arguments stored in an arrays of length 3 and 6, respectively. 6(CO1,3,4)
4. (A) Write a python script to perform moving average of five elements of an array  $X = [1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9 \ 10]$ . 5(CO1,3,4)  
(B) Write a python script to send distance measured using ultrasonic sensor only if it is greater than 10 cm and less than 30 cm to a Thingsboard server. 5(CO1,3,4)

5. (A) Write a python script to read an image frame from USB camera and detect face-mask is present or not. 5(CO1,3,4)
- (B) Compare I2C and UART protocol with examples. 5(CO4)
6. (A) Identify the challenges of implementing an IoT Use case for Underground Mining in India. How would you overcome those ? Justify. 7(CO1,5)
- (B) Suggest and explain a protocol to monitor weather in a desert environment using sensors with power constraints. 3(CO1)

